## Management of surgical patients receiving anticoagulation and antiplatelet agent

Estimating thromboembolic risk

| | High Risk | Intermediate risk | Low risk |
|---|---|---|---|
| AF | -CHA2DS2* score ≥3<br>- Recent (less than 3 months) stroke or transient ischaemic attack<br>- Rheumatic valvular heart disease | | -CHA2DS2* score of 1-2 |
| **Mechanical heart valves** | - Any mitral valve prosthesis<br>- Older (caged-ball) aortic valve prosthesis<br>- Recent (less than 6 months) stroke or transient ischaemic attack | -Bileaflet aortic valve prosthesis and one of the following:<br>● atrial fibrillation,<br>● prior stroke or transient ischaemic attack,<br>● hypertension,<br>● diabetes,<br>● congestive heart failure,<br>● age >75 years | -Bileaflet aortic valve prosthesis without atrial fibrillation and no other risk factors for stroke |
| **DVT and thromboembolism** | - Recent (less than 3 months) venous thromboembolism (VTE)<br>- Active cancer<br>- Severe thrombophilia (e.g. deficiency of protein C, protein S, antithrombin, antiphospholipid antibodies, multiple abnormalities) | -VTE 3-12 months prior<br>- Non-severe thrombophillic conditions (e.g. heterozygous factor V leiden mutation, prothrombin gene mutation)<br>- Recurrent VTE (treated within 6 months or palliative) | -Single VTE occurred >12 months ago and no other risk factors |

Government of Western Australia department of health.Guidelines for Anticoagulation using Warfarin.

Estimating procedural bleeding risk is based on type of surgery/intervention as well as patient comorbidities and medications that affect haemostasis. Refer to Table 1 for bleeding risk determination. Table 2 refers to the peri-operative management of warfarin.

**Table 1: Classification of elective surgical interventions according to bleeding risk**

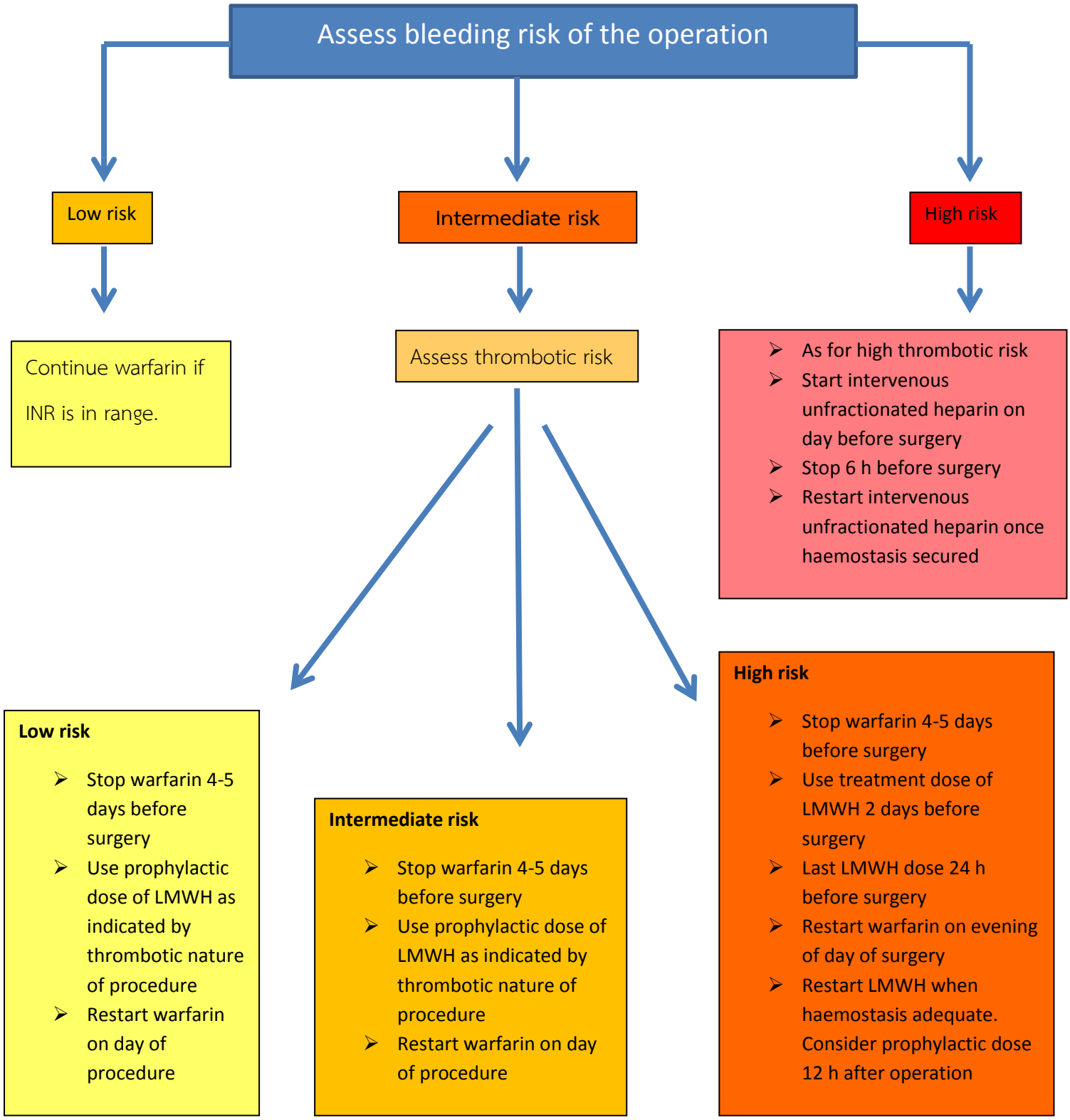
| **Interventions with minor bleeding risk** | **Interventions with low bleeding risk** | **Interventions with high bleeding risk** |
|---|---|---|
| Dental interventions<br>✓ Extraction of 1–3 teeth<br>✓ Paradontal surgery<br>✓ Incision of abscess<br>✓ Implant positioning | Non-coronary angiography (for coronary angiography and ACS: see Patients undergoing a planned invasive procedure, surgery or ablation section) | Complex endoscopy (e.g. polypectomy, ERCP with sphincterotomy etc.) |
| Cataract or glaucoma intervention | Prostate or bladder biopsy | Spinal or epidural anaesthesia; lumbar diagnostic puncture |
| Endoscopy without biopsy or resection | Endoscopy with biopsy | Liver biopsy/ Kidney biopsy/ Abdominal surgery/ Major orthopaedic surgery |
| Superficial surgery (e.g. abscess incision; small dermatologic excisions; . . .) | Electrophysiological study or catheter ablation (except complex procedures, see below)<br>Non-coronary angiography | Complex left-sided ablation (pulmonary vein isolation; some VT ablations) |

The 2018 European Heart Rhythm Association Practical Guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation.

**Table 2: Peri-operative management of warfarin**

| **Assess bleeding risk of surgery** |
|---|
| **Low bleeding risk** |
| Continue warfarin if INR is in range. Consider a lower peri-operative INR goal of 2. |
| **Intermediate to high bleeding risk** |
| Low thrombotic risk<br>● Stop warfarin 4-5 days before surgery if ordered<br>● No bridging anticoagulation<br>● No bridging anticoagulation if AF is the indication for warfarin unless patient has CHADS2 score of 5 or 6<br>● Restart warfarin on evening of day of surgery if appropriate |
| Intermediate thrombotic risk<br>● Stop warfarin 4-5 days before surgery if ordered<br>● Bridging anticoagulation to be considered if the thromboembolic risk clearly outweighs the increased bleeding risk from bridging<br>● Consider more conservative bridging strategies such as prophylactic dose of Low Molecular Weight Heparin (LMWH) 2-3 days before surgery<br>● Last prophylactic LMWH dose 12 hours before surgery if renal function is normal<br>● Restart warfarin on evening of day of surgery if appropriate<br>● Restart therapeutic bridging anticoagulation once haemostasis is adequate |
| High thrombotic risk<br>● Stop warfarin 4-5 days before surgery if ordered. It is important to consider an adequate period to withhold warfarin as the requirement to reverse with vitamin K prior to surgery will delay re-warfarinisation postsurgery (especially for patients with metallic heart valves).<br>● Consult haematology/cardiology to discuss bridging options. For example, patients with mechanical heart valves may require bridging with intravenous heparin infusion.<br>● If therapeutic dose of LMWH is ordered, start 2-3 days before surgery<br>● Commence when INR is subtherapeutic<br>● Last dose is 24 hours before procedure is commenced if renal function is normal |

- Consider switching to intravenous heparin if patient very high thrombotic risk – stop 4 hours prior to commencement of surgery
- Restart warfarin on evening of day of surgery if appropriate
- Restart bridging anticoagulation when haemostasis is adequate, for patients with high bleeding risk consider restarting LMWH at prophylactic dose or IV heparin without a bolus dose.




Government of Western Australia department of health.Guidelines for Anticoagulation using Warfarin.[online].Available from:

https://ww2.health.wa.gov.au/~/media/Files/Corporate/general%20documents/WATAG/Warfarin-guidelines-for-